# Critical SK Mixing: Quantitative Support-Profile Transfer and Physical Certificates

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M434

October 9, 2026

**Abstract**

The September 25 source manuscript asserts the critical worst-start mixing exponents 2/3 in rate-one-per-site time and 5/3 in attempted single-site updates. This report studies the deterministic transfer at the end of its support-profile proof, not an independent verification of the random critical profile. We extend the source's bounded-test extraction to arbitrary geometric level ratios, improving its explicit factor 49/200 to 7/20 with ratio five. A sharp two-moment truncation inequality gives an optimized density-energy envelope and a closed warm-up integral, with leading coefficient $1/(2\alpha)$ instead of $1/\alpha$ for a given logarithmic profile constant $\alpha$. The estimates allow a uniform additive profile defect and transfer directly to deterministic attempted updates by heat-bath positivity. Physical Dirichlet pencils make the all-support obligation explicit; small covariance or trial-space tests cannot discharge it. Freshly read original metric and spectral-limit arguments supply the norm and omitted-space distinctions. Exact finite checks supplement the proofs without certifying the critical disorder event. Contribution assessment: C2/5, explicit but limited deterministic constant and error refinements are proved, without a new critical profile, exponent, or global uniformity theorem.

## 1 Model, claims, and the actual interface

Let $\Omega = \{-1, 1\}^n$, and, conditional on a fixed symmetric matrix $J$, put

$$\mu(x) = Z^{-1} \exp\{\tfrac{1}{2} x^\top J x\}, \qquad x \in \Omega.$$

The source has independent $J_{ij} \sim N(0, 1/n)$, $i < j$, at inverse temperature one and zero external field. Adding a diagonal matrix changes the energy by a constant on the cube, so a GOE diagonal completion does not change $\mu$. All local propositions below hold for a fixed full-support probability $\mu$, not just Gaussian disorder. Write $Q_i f = \mathbb{E}_\mu[f \mid X_{-i}]$. The two clocks are

$$P = \frac{1}{n} \sum_i Q_i, \qquad L = \sum_i (Q_i - I) = n(P - I), \qquad \mathcal{D}(f) = \langle f, -Lf \rangle_\mu = \sum_i \mathbb{E}_\mu \operatorname{Var}_\mu(f \mid X_{-i}). \tag{1}$$

Thus $e^{tL}$ runs every site at rate one; $P^k$ makes exactly $k$ uniform-site attempts, including attempts that retain the spin. Each $Q_i$ is an orthogonal projection in $L^2(\mu)$. Consequently $P$ is self-adjoint with spectrum in $[0, 1]$.

The source [1] claims, for each fixed positive exponent slack, high-probability upper and lower bounds giving

$$t_{\mathrm{mix}}^{\mathrm{ct}}(1/4) = n^{2/3+o(1)}, \qquad t_{\mathrm{mix}}^{\mathrm{att}}(1/4) = n^{5/3+o(1)}$$

in probability over $J$. It supplies two upper-bound routes: a logarithmic support profile, and a spectral-cap route to a classical logarithmic Sobolev inequality. Its lower bound imports a critical-slowing companion. Those statements are source claims. Neither a publication date nor the finite computations here certify their full proofs.

### 1.1 All observables, rather than linear modes

For $f(x) = v \cdot x$,

$$\mathrm{Var}_\mu(f) = v^\top \mathrm{Cov}_\mu(X) v, \qquad \mathcal{D}(f) = \sum_i v_i^2 \mathbb{E}_\mu \mathrm{Var}(X_i \mid X_{-i}) \le \|v\|_2^2 .$$

A large covariance direction can therefore give a slow mode. Conversely, controlling these $n$ directions does not bound $\sup_f \mathrm{Var}_\mu(f)/\mathcal{D}(f)$, which ranges over a space of dimension $2^n - 1$. Positive semidefiniteness of covariance alone says even less. The source's stronger covariance-scale quantifiers, involving arbitrary diverging deterministic factors, must not be transferred to its all-observable inequalities, which are stated with fixed exponent slack.

### 1.2 A reusable, genuinely uniform hypothesis

For $\alpha > 0$ and $0 \le \eta < \alpha \log 2$, define the defective support-profile hypothesis

$$\mathcal{D}(f) \ge [\alpha \log(1/p) - \eta] \mu(f^2), \quad 0 \ne f \ge 0, \quad p = \mu(f > 0) \le \tfrac{1}{2}. \tag{2}$$

Every function in this display must be covered on the same deterministic environment. The exact source profile has $\eta = 0$ and, for fixed $\omega > 0$, a coefficient of order $c_\omega n^{-2/3-\omega}$. The report proves implications from (2); it does not manufacture this coefficient from finite covariance tests. The defect $\eta$ is a uniform form error, not an average error over a sample of test functions.

## 2 Geometric layers with an explicit cap cost

The last part of the source's `profile-entropy.tex` first establishes a bounded-test profile with cap $100p^{-2}$, and then removes the cap by dyadic layers. We keep that mechanism and compute its dependence on the layer ratio.

**Theorem 2.1** (Capped-to-full profile)**.** *Let a reversible finite-state Dirichlet form satisfy*

$$\mathcal{D}(h) \ge \alpha \log(1/p_h) - \eta$$

*for all nonnegative $h$ with $\mu h^2 = 1$, $0 < p_h = \mu(h > 0) \le 1/2$, and $\|h\|_\infty^2 \le B p_h^{-2}$. Assume $B > 1/2$, $\alpha > 0$, and $0 \le \eta < \alpha \log 2$. For any $1 < r < \sqrt{2B}$, every nonzero $f \ge 0$ of support mass $p \le 1/2$ satisfies*

$$\mathcal{D}(f) \ge \frac{r-1}{2r} \left(1 - \frac{r^2 p}{B}\right) [\alpha \log(1/p) - \eta] \, \mu f^2. \tag{3}$$

*In particular, (2) holds with*

$$(\alpha, \eta) \text{ replaced by } (\gamma\alpha, \gamma\eta), \qquad \gamma = \frac{r-1}{2r}\left(1 - \frac{r^2}{2B}\right).$$

*Proof.* For $j \in \mathbb{Z}$, define

$$g_j = \min\{(f - r^j)_+, (r-1)r^j\}, \qquad A_j = \mu g_j^2, \quad p_j = \mu(f > r^j).$$

If $u = r^m x > 0$, $1 \le x \le r$, summing the geometric tail gives

$$\sum_j g_j(u)^2 = r^{2m}\left(\frac{r-1}{r+1} + (x-1)^2\right).$$

Dividing by $u^2$, the minimum occurs at $x = 2r/(r+1)$, with value $(r-1)/(2r)$. Also $\sum_j g_j(u) = u$, so $\sum_j g_j(u)^2 \le u^2$. Hence

$$\frac{r-1}{2r}\mu f^2 \le S := \sum_j A_j \le \mu f^2. \tag{4}$$

For $u \ge v$, all increments $g_j(u) - g_j(v)$ are nonnegative and sum to $u - v$. Thus their squared sum is at most $(u-v)^2$. The edge representation of a reversible Dirichlet form implies $\sum_j \mathcal{D}(g_j) \le \mathcal{D}(f)$. Infinite sums cause no limiting difficulty: all terms are nonnegative and the displayed pointwise geometric sums are finite.

Discard zero layers and layers with

$$A_j < (r-1)^2 r^{2j} p_j^2 / B.$$

For the latter,

$$A_{j-1} \ge (r-1)^2 r^{2j-2} p_j, \qquad A_j \le \frac{r^2 p_j}{B} A_{j-1} \le \frac{r^2 p}{B} A_{j-1}.$$

Summation, including overlaps only through the injective shift $j \mapsto j-1$, gives $S_{\text{kept}} \ge (1 - r^2 p/B)S$. Every retained $g_j/\sqrt{A_j}$ meets the bounded-test cap. Since $p_j \le p$ and $\alpha \log(1/p) - \eta > 0$,

$$\mathcal{D}(f) \ge \sum_{\text{kept}} \mathcal{D}(g_j) \ge [\alpha \log(1/p) - \eta] S_{\text{kept}}.$$

Apply (4). □

**Corollary 2.2** (The source cap $B = 100$)**.** *The dyadic choice $r = 2$ yields $49/200$, as in the source. The rational choice $r = 5$ yields $7/20$, an improvement by the factor $10/7$, under the same capped-test hypothesis. Within the displayed one-parameter family, the maximizing ratio is the unique $r > 1$ satisfying $r^2(2r-1) = 2B$.*

*Proof.* Substitute the two ratios. Differentiating $\gamma$ gives $\gamma'(r) = 1/(2r^2) - (2r-1)/(4B)$. The cubic $r^2(2r-1)$ increases strictly for $r > 1$; its values at 1 and $\sqrt{2B}$ bracket $2B$. □

This is a constant optimization of classical layer extraction, not a new critical estimate. The maximizing ratio need not be globally optimal among different decompositions or different proofs. Importantly, $\eta/\alpha$ is unchanged by the uniform extraction.

## 3 Sharp density truncation and its optimized envelope

**Lemma 3.1** (Two-moment truncation)**.** *On any probability space, let $g \ge 0$, $\mu g = 1$, and $\mu g^2 = V < \infty$. For $0 \le a \le V$,*

$$\mu(g-a)_+^2 \ge V(1 - a/V)^2. \tag{5}$$

*The bound is sharp among all laws with these two moments.*

*Proof.* For $x \geq 0$ the quadratic minorant

$$(x-a)_+^2 \geq (1-a^2/V^2)x^2 - 2a(1-a/V)x$$

holds. For $x \leq a$, its right side is nonpositive: the nonzero root of that quadratic is $2a/(1+a/V) \geq a$, with the endpoint $a = V$ immediate. For $x \geq a$, the difference is exactly $a^2(x-V)^2/V^2$. Integrating proves (5). Equality holds for a random variable equal to $V$ on a set of probability $1/V$ and zero elsewhere. This extremal law need not be representable by a particular fixed Gibbs measure. □

Set $\beta = \eta/\alpha \in [0, \log 2)$; this symbol is a defect ratio, not the SK inverse temperature, which remains one.

**Theorem 3.2** (Optimized profile energy)**.** *Assume* (2)*. For $g \geq 0$, $\mu g = 1$, $V = \mu g^2 \geq 8$, let $a_\beta(V)$ be the unique solution in $(2, V)$ of*

$$V = a\{1 + 2(\log a - \beta)\}, \qquad t_\beta(V) = V/a_\beta(V). \tag{6}$$

*Then*

$$\mathcal{D}(g) \geq \alpha \Phi_\beta(V), \qquad \Phi_\beta(V) = \frac{V(t_\beta(V)-1)^3}{2t_\beta(V)^2}. \tag{7}$$

*This is the maximum of the lower bounds obtained from* (5)*, Markov's inequality, and one cutoff $a \in [2, V]$.*

*Proof.* The cutoff is 1-Lipschitz, so $\mathcal{D}(g) \geq \mathcal{D}((g-a)_+)$. Its support has mass at most $1/a \leq 1/2$. For $2 \leq a < V$, the cutoff is nonzero by (5), and

$$\mathcal{D}(g) \geq \alpha V(1-a/V)^2(\log a - \beta).$$

The derivative of the last factorized expression, apart from positive factors in the interior, has the sign of $V/a - 1 - 2(\log a - \beta)$. Its unique zero is (6). The function $a[1 + 2(\log a - \beta)]$ has positive derivative $3 + 2(\log a - \beta)$ on $a \geq 2$. At $a = 2$ its value is at most $2 + 4\log 2 < 8$, whereas at $a = V$ it exceeds $V$. Thus the maximizer lies in $(2, V)$. Finally, $\log a - \beta = (t-1)/2$ and $1 - a/V = (t-1)/t$, which give (7). □

For $\eta = 0$, the source uses $a = V/4$, a weaker norm estimate $\mu(g-a)_+^2 \geq V/2$, and therefore $\mathcal{D}(g) \geq \alpha V \log(V/4)/2$ for $V \geq 8$. Even retaining its cutoff, (5) improves $1/2$ to $9/16$; optimizing improves it further. The scalar moment inequality is sharp, but the combined energy envelope need not be sharp for a given Markov chain: Markov's support bound and the Dirichlet contraction can have slack.

## 4 An explicit warm-up time and the exact clock transfer

**Lemma 4.1** (Gap from a defective profile)**.** *Under* (2)*,*

$$\mathcal{D}(f) \geq \lambda \operatorname{Var}_\mu(f) \quad \textit{for every real } f, \qquad \lambda = \alpha \log 2 - \eta > 0.$$

*Proof.* Choose a median $m$ of $f$. Both $f_+ = (f-m)_+$ and $f_- = (m-f)_+$ have support mass at most $1/2$. For each pair of states the squared increment of $f$ is at least the sum of the squared increments of these two parts. Hence

$$\mathcal{D}(f) \geq \mathcal{D}(f_+) + \mathcal{D}(f_-) \geq \lambda \mu[(f-m)^2] \geq \lambda \operatorname{Var}_\mu(f).$$

Zero parts contribute zero. This proof includes every observable, not just a coordinate or a covariance eigenvector. □

Define the increasing function, for $t > 1$,

$$F(t) = \log(t-1) - \frac{4}{t-1} - \frac{3}{2(t-1)^2}, \qquad F'(t) = \frac{t(t+2)}{(t-1)^3} > 0. \tag{8}$$

For $M \geq R \geq 8$, put

$$T_\beta(M,R) = \frac{F(t_\beta(M)) - F(t_\beta(R))}{2\alpha}. \tag{9}$$

**Theorem 4.2** (Profile-to-mixing certificate)**.** *Assume (2), and fix $R \geq 8$, $0 < \varepsilon < 1/2$. For an initial density $g_0 \geq 0$, $\mu g_0 = 1$, put $M = \max\{R, \mu g_0^2\}$. At all times*

$$t \geq T_\beta(M,R) + \frac{1}{2(\alpha \log 2 - \eta)} \log \frac{R-1}{4\varepsilon^2}, \tag{10}$$

*the continuous-time distribution has total variation distance at most $\varepsilon$ from $\mu$. The same conclusion for exactly $k$ heat-bath attempts holds whenever $k$ is at least the ceiling of $n$ times the right side of (10). For worst starts one may take $M = \max\{R, 1/\mu_{\min}\}$.*

*Proof.* By reversibility $g_t = e^{tL} g_0$, and with $V_t = \mu g_t^2$,

$$V_t' = -2\mathcal{D}(g_t) \leq -2\alpha \Phi_\beta(V_t) \quad (V_t \geq R).$$

The time needed to reach $R$ is at most $\int_R^M dV/(2\alpha\Phi_\beta(V))$. Equation (6) is equivalent to

$$V = e^\beta t e^{(t-1)/2}, \qquad \frac{dV}{V} = \frac{t+2}{2t}\, dt.$$

Substitution gives exactly (9). Subsequently Lemma 4.1 gives $V_{T_\beta + s} - 1 \leq (R-1)e^{-2\lambda s}$. Cauchy–Schwarz yields $\mathrm{TV}(g\mu, \mu) \leq \frac{1}{2} \|g - 1\|_{L^2(\mu)}$. The same estimates apply if $V_0 \leq R$, with zero warm-up time.

For the discrete assertion, diagonalize the positive contraction $P$ in $L^2(\mu)$. For each eigenvalue $\theta \in [0,1]$ and integer $k \geq 1$,

$$\theta^{2k} \leq e^{-2k(1-\theta)}.$$

Consequently, for every $h$,

$$\left\| P^k h \right\|_{L^2(\mu)}^2 \leq \left\| e^{(k/n)L} h \right\|_{L^2(\mu)}^2. \tag{11}$$

Apply this to $h = g_0 - 1$ and the already proved continuous-time $L^2$ estimate. Finally a point mass at $x$ has density $\mathbf{1}_{\{x\}}/\mu(x)$ and squared norm $1/\mu(x)$. □

### 4.1 What changes in the source bound

For fixed $\beta < \log 2$ and fixed $R$,

$$T_\beta(M,R) = \frac{1}{2\alpha} \log\log M + O_{\beta,R}(\alpha^{-1}) \qquad (M \longrightarrow \infty). \tag{12}$$

Indeed $t + 2\log t = 2\log M - 2\beta + 1$, so $t/(2\log M) \to 1$, and $F(t) = \log t + o(1)$. The source's fixed-cutoff calculation gives $\alpha^{-1} \log\{\log(M/4)/\log 2\}$, with leading coefficient $1/\alpha$. The improvement is a comparison of deterministic certificates with the same supplied $\alpha$, not a sharp mixing constant for critical SK.

The source's direct one-step profile argument uses $P^2 \preceq P$ and loses another factor two in discrete time. Equation (11) avoids that loss. This positive-spectrum comparison is classical and is already used in the source's logarithmic-Sobolev route; its reuse here is not credited as a new spectral principle. It does not assert that discrete total variation is pointwise bounded by continuous total variation. The comparison is specifically in $L^2$.

*Remark* 4.3 (A simpler, weaker optimization). Replacing Lemma 3.1 by $\mu(g-a)_+^2 \geq V - 2a$, and taking $\eta = 0$, gives $\mathcal{D}(g)/(\alpha V) \geq (1 - 2a/V)\log a$. Its interior optimizer satisfies $V = 2qe^{q-1}$, $a = e^{q-1}$, and the optimized factor is $(q-1)^2/q$. The condition $a \geq 2$ is $V \geq 4(1+\log 2)$. Since $d\log V = (q+1)dq/q$, the warm integral has primitive $\{\log(q-1) - 2/(q-1)\}/(2\alpha)$. This independently checks the improved leading coefficient. The quadratic certificate dominates this linear one for every common cutoff, and optimizes over the larger interval $2 \leq a \leq V$. No historical novelty is asserted for either one-variable optimization.

### 4.2 Limits and a necessary positivity gate

For the two-state deterministic swap, $P$ has eigenvalue $-1$. The attempted-update chain never converges, while $e^{t(P-I)}$ does converge. Thus reversibility alone cannot justify (11). Heat-bath positivity is the essential additional gate.

If $\eta$ approaches $\alpha\log 2$, the gap tail in (10) diverges. A defect small in absolute units is not sufficient when the target profile coefficient itself vanishes like $n^{-2/3-\omega}$. One needs $\eta/\alpha$ bounded strictly below $\log 2$. The local estimates are valid simultaneously for a whole family of environments only when that relative-error condition and the full profile are simultaneous.

## 5 Physical Dirichlet pencils and certified defects

This section specializes the freshly read original Ram physical-metric arguments [6, 7]. It is a supporting certification framework, not an additional critical-mixing theorem.

For an unordered spin-flip edge $\{x, y\}$, the rate from $x$ to $y$ is $\mu(y)/(\mu(x)+\mu(y))$. Therefore its conductance is

$$c_{xy} = \frac{\mu(x)\mu(y)}{\mu(x)+\mu(y)}, \qquad \mathcal{D}(f) = \sum_{\{x,y\}} c_{xy}|f(x) - f(y)|^2.$$

Let $K$ be the weighted graph Laplacian and $G = \mathrm{diag}(\mu(x))$. Then $\mathcal{D}(f) = f^*Kf$, $\mu|f|^2 = f^*Gf$. For a nonempty subset $A$, extending $f_A$ by zero gives

$$\lambda_D(A) = \min_{v\neq 0} \frac{v^*K_{AA}v}{v^*G_{AA}v}. \tag{13}$$

The diagonal of $K_{AA}$ retains all conductances to $A^c$; it is not the Laplacian of the induced subgraph with those edges deleted. This is a Dirichlet, not a reflecting, support restriction.

**Proposition 5.1** (All-support equivalence and form error)**.** *The full profile* (2) *is equivalent to*

$$K_{AA} \succeq [\alpha\log(1/\mu(A)) - \eta]G_{AA} \quad \textit{for every nonempty } A,\ \mu(A) \leq 1/2. \tag{14}$$

*In particular, suppose certified matrices* $\widehat{K}_{AA}$ *satisfy, for every such* $A$*,*

$$\widehat{K}_{AA} \succeq \alpha\log(1/\mu(A))G_{AA}, \qquad |v^*(\widehat{K}_{AA} - K_{AA})v| \leq \eta\, v^*G_{AA}v \quad \textit{for all } v.$$

*Then* (2) *and Theorem 4.2 hold.*

*Proof.* For any real or complex $v$, entrywise absolute values do not increase the edge energy. Apply (2) to $|v|$; its actual support has mass at most $\mu(A)$. This proves (14). Conversely choose $A = \{f > 0\}$. The error conclusion follows by subtracting the asserted uniform quadratic-form error. No passage from a sampled vector list to all vectors is used. □

Under an invertible synthesis $v = Tc$ on a fixed support,

$$G_T = T^* G_{AA} T, \quad K_T = T^* K_{AA} T, \quad \|v\|^2_{L^2(\mu)} = c^* G_T c.$$

The pencil inequality is preserved by congruence, not by ignoring $G_T$. If $T$ has fewer columns than $|A|$, positivity of the compressed pencil controls only its range. Its minimum Rayleigh value is an upper bound on $\lambda_D(A)$, not a lower bound. This is the semibounded variational distinction proved in the original spectral-limit chapter, rather than an application of its indefinite spectral-pollution example to a positive Laplacian.

### 5.1 Worked two-spin support certificate

For independent fair spins with $n = 2$, order the states cyclically around the square. Then

$$G = \frac{1}{4} I_4, \qquad K = \frac{1}{8} \begin{pmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}.$$

Singletons have $\lambda_D = 1$. Adjacent pairs have $\lambda_D = 1/2$, and opposite pairs have $\lambda_D = 1$. These are all ten nonempty subsets with mass at most $1/2$. Thus (2) holds with $\alpha = 1/2, \eta = 0$: $\frac{1}{2}\log 4 < 1$ and $\frac{1}{2}\log 2 < 1/2$. If the two boundary edges of an adjacent pair were removed, the resulting 2-vertex reflecting Laplacian would have a zero eigenvalue. That matrix would answer a different question. This exact independent-spin example is not a Gaussian critical-disorder sample and does not test any large-$n$ event.

## 6 A finite numerical budget with exact enclosures

Suppose a full profile has been supplied with $\alpha = 1/2, \eta = 0$, and an initial density has $V_0 \le 2^{20}$. Take $R = 8$ and $\varepsilon = 1/4$. The owned rational-arithmetic checker encloses the two warm-up bounds and their common gap tail as follows:

| Quantity | Lower enclosure | Upper enclosure |
|---|---|---|
| Optimized $T_0(2^{20}, 8)$ | 4.604107 | 4.604108 |
| Source fixed-cutoff warm-up | 5.780743 | 5.780744 |
| Common gap tail $\log 28/\log 2$ | 4.807354 | 4.807355 |

Each finite decimal endpoint denotes an exact rational, rounded outwards. In particular 9.411463 is a certified continuous-time sufficient bound in this hypothetical example. The corresponding attempted-update certificate is $\lceil 9.411463\, n \rceil$. The example supplies no assertion that a particular critical SK realization has $\alpha = 1/2$, or that the independent two-spin example has $V_0 = 2^{20}$.

For reproducibility, if $1 \le x \le 2$ and $z = (x-1)/(x+1)$, the checker uses

$$2 \sum_{j=0}^{m-1} \frac{z^{2j+1}}{2j+1} \le \log x \le 2 \sum_{j=0}^{m-1} \frac{z^{2j+1}}{2j+1} + \frac{2z^{2m+1}}{(2m+1)(1-z^2)}.$$

These follow by integrating the geometric series for $2/(1-z^2)$. Power-of-two reduction handles other positive rational arguments. With $m = 40$, rational bisection of (6) gives upper and lower root endpoints. Monotonicity of $F$ then encloses (9); floating-point evaluations are used only for displaying the already certified rational intervals.

# 7 Source architecture and the critical obligations

The entire source architecture was mapped, but most of its 14,823 lines were not literally read. The selected route includes the complete entropy-localization/profile and final dynamics modules. The following map distinguishes those proof readings from imported modules. *Mapped* means inclusion and structural headings were inspected, not a proof audit.

| Source module or group | Role and inspection boundary |
|---|---|
| Introduction, model, accepted inputs, proof overview | Read in full. Define the clocks, two upper routes, companion inputs, and the distinction between covariance and all-function claims. |
| Profile spectral, roots, matrices, orbits | Mapped, not literally proof-read. Edge resolvents, root conditioning, matrix control and rotation arguments remain imported. |
| Profile local, amplification, positive | Mapped, not literally proof-read. Local spin comparisons, backward amplification and positive-observation estimates are the principal unverified inputs to the read entropy module. |
| Profile entropy | All 694 lines read. Observation identities, three support-size regimes, and the final bounded-test extraction form the attachment point of Theorem 2.1. |
| Cap static | Lines 1–118 read. Observation precision, spectral caps and bottom scales checked as conventions. Remaining spherical comparisons and tails imported. |
| Cap multiscale | Lines 1–124 read. Radial/transverse invariant and one-step hypotheses inspected; the subsequent propagation proof imported. |
| Cap classical | Mapped, not literally proof-read. Classical auxiliary input retained as part of the cap route. |
| Cap initialization | Lines 976–1049 read: final true-potential curvature theorem and assembly. The initialization, auxiliary removal and intermediate support constructions remain imported. |
| Cap entropy | Lines 1–151 and 851–909 read: capacity transfer and final LSI assembly. The intervening entropy argument remains imported. |
| Dynamics | All 310 lines read. Profile warm-up, clocks, LSI alternative, lower bounds and entropy production were inspected. |
| Paper, bibliography, preamble, README | Paper and bibliography read in full. Preamble and source README mapped only; neither counts as scientific proof reading. |

## 7.1 Observation law and square reweighting

For a prior $\mu_{J,h}$, a Gaussian observation with precision $B \succeq 0$ is

$$Y = BX + B^{1/2}G, \qquad \mu(X = x \mid Y = y) \;\propto\; \exp\{\tfrac{1}{2}x^\top (J - B)x + (h + y)^\top x\}.$$

For singular $B$, the Gaussian likelihood is on $\operatorname{ran} B$. The subtraction of $B$ from the interaction is essential; diagonal precision has a constant quadratic term on the cube, whereas a general spectral cap does not. The source's ordinary observation law and its test-dependent square-reweighted law are not interchangeable.

In the read entropy route, a normalized test induces $\nu = f^2\mu$, a posterior weight $w_b = \mu_b f^2$, and $H_b = \mathbb{E}_\nu \operatorname{KL}(\nu_b \| \mu_b)$. The source uses a conditional mean displacement $a_b$ and a precision rate $\Lambda_b$ to

obtain

$$H_b' = -\tfrac{1}{2}\mathbb{E}_\nu \left\| \Lambda_b^{1/2} a_b \right\|^2 .$$

Its backward mean identity and terminal product estimates are applied under this actual reweighted law. Rare path probabilities under the ordinary law need likelihood and entropy conversion, not direct substitution. No local optimization here removes that requirement.

### 7.2 Why the entropy module is not just an ODE

For support entropy $h = \log(1/p)$, the source divides into small, moderate, and macroscopic $h$. The moderate regime uses the critical observation scale $(h/n)^{2/3}$, early likelihood control, fixed-ratio propagation, a later drift coefficient close to $1/b$, and a terminal product inequality with a nontrivial random-field cost. The small regime instead retains variance and integrates covariance bounds through the bottom scale. In the macroscopic regime the proof takes uniform limits of entropy losses; a binary entropy conversion and root-tail estimates force the limiting loss to vanish, leading to a contradiction at the product endpoint.

These are distinct source arguments. Their good environment is selected before the test function, with integrated exceptional events and entropy-bin control. Replacing them by a separate high-probability event for each $f$ would not prove the simultaneous hypothesis used in this report. The unrestricted profile then follows by the read layer extraction. Our change to the ratio does not change any earlier likelihood, root or exceptional-event estimate.

### 7.3 What cap curvature actually says

The cap convention is

$$d = 2I - J, \quad B_p = (p^2 - d)_+, \quad J_p = J - B_p, \quad P_p = \mathbf{1}_{\{d \le p^2\}}, \quad k_p = np^3.$$

The true cap potential is $\phi_p(h) = \log Z_{J_p}(h)$, and $\nabla^2 \phi_p(h) = \mathrm{Cov}_{\mu_{J_p,h}}(X)$. The read invariant concerns the compression to $H_p = \operatorname{ran} P_p$, on an annulus separated from zero. Its scaled difference from a spherical resolvent has a negative radial component and controlled transverse and mixed components. This is stronger than positivity of a covariance matrix and different from an all-field ambient covariance bound.

The source's final curvature theorem gives a sharp $(1+\epsilon)/p^2$ projected bound on a thin saddle-parameter annulus, and a coarser $C/p^2$ bound on a prescribed field ball, down to $np^3 \asymp \sqrt{\log n}$. Extending from geometric scales to every intermediate scale requires the source's deterministic evaluation lemmas. Its true potential must also replace the auxiliary initialization after a fixed number of levels. Fixed field ranges and tolerances precede $n \to \infty$. No finite pencil calculation here establishes that replacement, the unused-Haar conditional independence, or the simultaneous all-field event.

### 7.4 Conditional application to the asserted critical profile

If the source's capped-test profile holds on events $\mathcal{E}_n$ with probability tending to one, then Theorem 2.1 holds on exactly those events. There is no extra probability union over observables. If $\alpha_n \ge c_\omega n^{-2/3-\omega}$ after extraction, $\eta_n = 0$, and $\|J\| \le 3$, then

$$\mu_{\min} \ge \exp\{-(3 + \log 2)n\}.$$

This follows by bounding the range of $\frac{1}{2}x^\top Jx$ by $3n$ and the partition sum by $2^n$ times its largest summand. Theorem 4.2 therefore gives $O_\omega(n^{2/3+\omega} \log n)$ rate-one time for fixed accuracy, and $n$

times that many attempts. Choosing a smaller fixed $\omega$ absorbs the logarithm into any prescribed exponent slack. This recovers only the source's upper exponent, conditionally on its profile. The companion lower bound, worst-start interpretation, and disorder probability statements are not proved by this transfer.

# 8 Original-proof connection, checks, and current status

The original Volume 21 physical-metrics chapter was freshly read in full (210 lines). Its proofs establish the transported Gram norm, the orthogonal decomposition of a full residual into projected and omitted pieces, and the need for relative rather than merely entrywise metric accuracy. These justify keeping $G_{AA}$ and all boundary conductances in Proposition 5.1. A nonorthogonal representation does not turn a small coordinate residual into a physical bound.

The original spectral-limits chapter was also freshly read in full (242 lines). It explicitly separates graph-core density, strong resolvent convergence, norm-resolvent control, and semibounded variational approximation. Its latter statement is the relevant one here: minima on trial subspaces approach a full infimum from above under sufficient form approximation. It does not produce a lower gap certificate from a chosen finite span. The catalogue's core-pattern report was retained as a locator and left unread; no result is attributed to a fresh reading of it.

## 8.1 Executable scope

The independent file `M434-exact-checks.py` uses rational arithmetic, with outward logarithm enclosures where needed. It passes 2,593 parameter cases and 5,180 assertions:

| Check family | Cases | Assertions |
|---|---|---|
| Pointwise quadratic minorant | 1395 | 2550 |
| Sharp two-point moment laws | 54 | 162 |
| Layer mass and ratio optimization | 67 | 70 |
| Layer edge-energy decomposition | 693 | 1386 |
| Optimized roots and primitive | 157 | 374 |
| Every half-mass support, uniform cubes $n = 1, 2, 3$ | 174 | 530 |
| Physical congruence and boundary distinction | 52 | 104 |
| Outward warm-up comparison | 1 | 4 |

The cube checks enumerate respectively 2, 10, and 162 support sets and certify positive exact LDL pivots after using an upper logarithm enclosure. They do not sample Gaussian disorder or validate any asymptotic high-probability assertion. Cases count input fixtures; assertions count explicitly executed checks, including matrix pivots. Reviewers' separate tests are excluded.

The literal public read scope is 2,611 of 14,823 lines across the 22 mapped cached text files; this includes 391 bibliography lines and 36 main-file lines. The two original private chapters contribute 452 freshly read lines. The audit ledger records per-file pre-read and post-read SHA256 values, actual display intervals and unread complements. Inclusion in the source tree, a heading match, or a citation is not counted as reading. Captured-tree identity against the October 8 snapshot is provenance only, not live-branch status or mathematical validation.

## 8.2 Primary literature refreshed on October 9, 2026

Goel–Montenegro–Tetali [2] develop the spectral-profile method for finite chains. Their primary text's definitions, truncation lemma and continuous-time comparison were inspected in the opening

two sections. The present work uses that classical framework; its scalar and layer refinements are source-specific calculations, not a claim to invent profile mixing bounds.

The primary abstracts and version records for Wang [3] and Boban–Li–Oveis Gharan [4] describe, respectively, optimal-order mixing for fixed inverse temperature below $1/2$, uniformly in external fields, and polynomial mixing up to $1/2 + \epsilon$ for an absolute positive $\epsilon$. Neither cited scope is a critical inverse-temperature-one worst-start theorem. Du–Huang [5] discuss critical free-energy fluctuations and overlap scale; those are static results, not a substitute for a full support profile. These three current papers were checked at abstract/version level, not independently proof-audited. No inference that the entire current literature has been exhausted is intended.

## 9 Advantages, costs, extensions, and conclusion

The useful increment is an explicit deterministic accounting of constants at a real source interface. Ratio-five extraction retains more layer mass under the same test cap. Sharp density truncation yields a stronger energy lower bound, an elementary monotone root equation, and an exact antiderivative for the warm-up budget. The defect ratio survives the extraction unchanged, making a physical-form error budget directly visible in the eventual mixing estimate. Deterministic attempts then use their actual positive-spectrum clock rather than a weaker one-step inequality.

The costs are equally important. All-support certification is combinatorially large, the critical $\alpha_n$ remains a difficult source input, and a fixed absolute error need not be small compared with it. Scalar sharpness does not make the Markov-chain estimate optimal. An improved leading coefficient in a supplied-profile calculation is not a cutoff theorem, a sharp leading SK mixing constant, or a better $2/3$ exponent. No defect in the source's retained dyadic extraction or fixed-cutoff argument was established; these are valid, nonoptimized estimates.

Feasible next steps are to use a scale-dependent certified profile in place of $\alpha \log(1/p) - \eta$, optimize the joining level $R$ against a separately available full gap, or choose the geometric ratio using the actual cap $B$ and support mass. Each would require a stated uniform input, not sampled covariance evidence. A practical large-$n$ certificate would additionally need structure that controls every support set without enumerating them. That remains unprovided by the finite examples.

The report consequently retains applicability L3 while assigning contribution C2: the proved refinements are useful and quantitative, but modest applications of classical truncation, layer decomposition and spectral calculus. The score is based on this limited increment, not on a rule that only a new global theorem could merit a higher score.

## References


[1] OpenAI, *Critical mixing in the Sherrington–Kirkpatrick model*, September 25, 2026. Cached manuscript, especially `profile-entropy.tex` and `dynamics.tex`. Captured-source identity and scoped proof readings accompany this report.

[2] S. Goel, R. Montenegro and P. Tetali, *Mixing Time Bounds via the Spectral Profile*, Electronic Journal of Probability **11** (2006), 1–26. arXiv:math/0505690; doi:10.1214/EJP.v11-300.

[3] S. Wang, *Optimal Mixing of Glauber Dynamics for the Sherrington–Kirkpatrick Model at $\beta < 1/2$*, arXiv:2608.22159v2, August 28, 2026. Primary abstract and version record checked.

[4] M. Boban, A. Li and S. Oveis Gharan, *Rank-1-perturbed trickledown theorems: Mixing time of Glauber dynamics for the Sherrington–Kirkpatrick model up to* $\beta \leq 1/2 + \varepsilon$, arXiv:2609.13138v1, September 11, 2026. Primary abstract and version record checked.

[5] H. Du and B. Huang, *Fluctuations of the Sherrington–Kirkpatrick free energy at critical temperature*, arXiv:2607.02172v2, July 8, 2026. Primary abstract and version record checked.

[6] M. Subbarao, Ram Calculus book corpus, Volume 21, original chapter file `ch01d-physical-metrics.tex`, lines 1–210. Confidential local manuscript; fresh literal reading, not uploaded or queried online.

[7] M. Subbarao, Ram Calculus book corpus, Volume 21, original chapter file `ch01f-spectral-limits.tex`, lines 1–242. Confidential local manuscript; fresh literal reading, not uploaded or queried online.